PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 6

One signal, two views

Time series in the time and frequency domains

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Where we left off

From Lectures 4 and 5

- Interpolation flattens fast oscillations and can shift phases
- Padding is extrapolation
- Differentiation amplifies fast fluctuations; integration accumulates slow ones
- Successive values not independent: standard errors and p-values do not hold; the shuffle test with the lag-1 autocorrelation r_1 checks this

Today

- What a time series is, and what we can read from it in time
- The same series as a sum of oscillations: Fourier analysis
- What sampling, record length and windows do to a spectrum
- When the oscillations change in time

Three views of a time series

View	Question	Tools	Part
Time domain	How does the value change from one moment to the next?	Plot in order; running mean and spread; autocorrelation	A
Frequency domain	Which oscillations is the series made of, and how strong are they?	Discrete Fourier transform (FFT); spectrum	B
Time and frequency	Do these oscillations change in time?	Short-time Fourier transform	C

The views are not competitors

The frequency domain contains the same information as the time series – arranged differently. Some questions are easy in one view and hard in the other.

PART A

The time domain

What is a time series, and what can we read from it directly?

What is a time series?

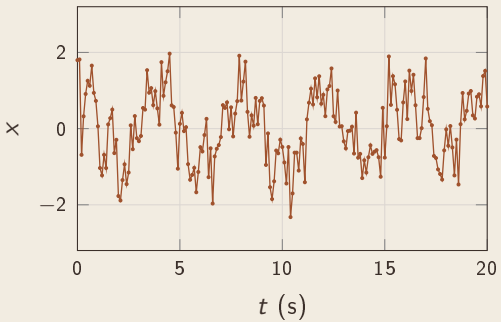
A time series

Values of one quantity at successive times, kept in time order:

$x_n = x(t_n), \quad t_n = n\,\Delta t \quad (n = 0, \ldots, N-1)$

- Δt: sampling interval; $f_s = 1/\Delta t$: sampling rate
- N values; duration $T = N\Delta t$
- Today: uniform sampling (irregular: Lecture 4)

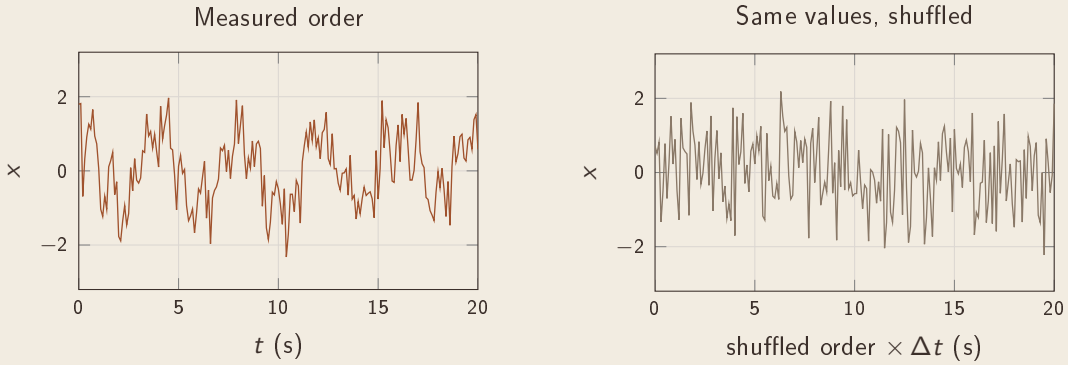
Running example: first 20 s shown



Running example (simulated): two oscillations, amplitude 1 at 0.25 Hz (period 4 s) and 0.5 at 1.1 Hz (period 0.91 s), plus independent Gaussian noise ($\sigma = 0.5$); $\Delta t = 0.1$ s, $T = 60$ s.

Two numbers set the limits of everything that follows: Δt and T.

The order carries the information

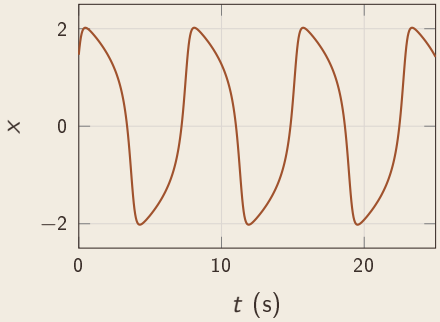


Same values, shuffled: a **surrogate** (Lecture 5). Same mean, s and histogram. Everything that distinguishes the two lies in the order: rhythms, trends, memory.

What we can read directly in time

Problem: what can we learn from one series without any transform?

van der Pol oscillator, $\mu = 2$ (RK4)

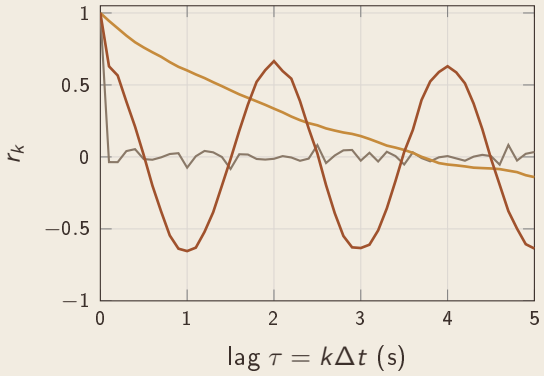


$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$, sampled every $\Delta t = 0.05\,\text{s}$

Feature	How, and the value here
Amplitude	largest $\lvert x \rvert$: 2.02
Period	time between upward zero crossings: 7.630 s, the same for all 51 periods of a 400 s run (plot: first 25 s)
Shape	slow stretches, fast jumps: *not* a cosine
Slow changes, sudden events	mean over a sliding window of several periods (running mean); steps or spikes by eye. Here: none
Memory	autocorrelation (next slide)

Autocorrelation: how long does the series remember?

Problem: how long does a value influence the ones that follow?



$$r_k = \frac{\sum_{n=0}^{N-1-k}(x_n - \bar{x})(x_{n+k} - \bar{x})}{\sum_{n=0}^{N-1}(x_n - \bar{x})^2}$$

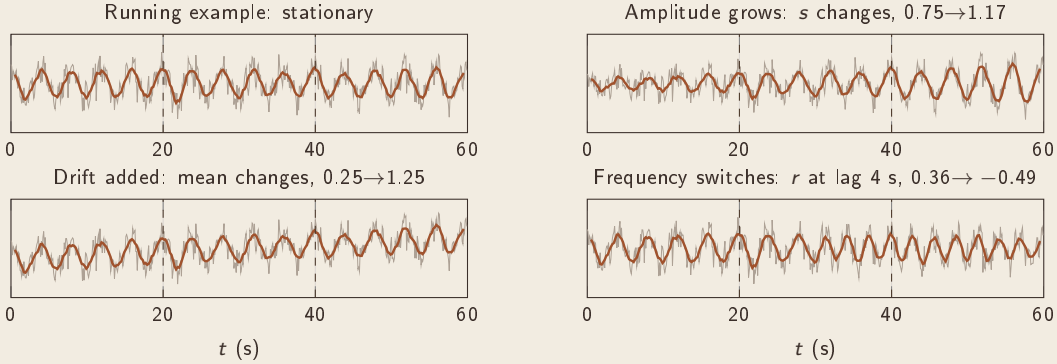
r_k generalises the lag-1 autocorrelation r_1 of Lecture 5 to a shift of k samples.

Three series (600 values, $\Delta t = 0.1$ s; colours as in the plot):

- **independent noise**: $r_k \approx 0$ for $k \geq 1$
- **with memory**: $x_n = a\,x_{n-1} + \varepsilon_n$, $a = 0.95$, independent ε_n: $r_k \approx a^k = e^{-\tau/\tau_c}$, correlation time $\tau_c = 1.95$ s
- **oscillation (period 2 s) + noise**: r_k oscillates with the same period

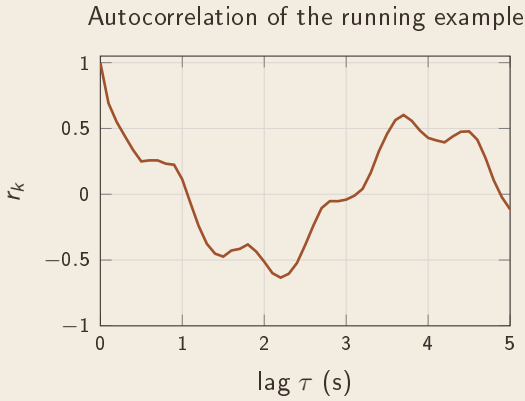
Stationary or not?

A series is **stationary** if its mean, standard deviation and autocorrelation do not change in time. A change of frequency or phase changes the autocorrelation even when the mean and s stay the same.



Grey: series; red: mean over 1 s, showing the slow rhythm; dashed: 20-s windows. Switch: 0.25→0.35 Hz at 30 s. A parameter varying explicitly in time (nonautonomous, Lecture 2) generally makes a series nonstationary.

What the time domain does not separate



The running example holds two rhythms, periods 4 s and 0.91 s, plus noise. In time:

- the slow rhythm dominates the series and its autocorrelation
- the fast one shows only as small ripples
- their sizes cannot be read off

Next

Write the series as a **sum of oscillations**, and measure each one separately.

PART B

The frequency domain

The same series as a sum of oscillations.

One oscillation: frequency, amplitude, phase

Problem: how do we describe one oscillation, and then a whole record as a sum of them?

$$x(t) = A\cos(2\pi f t + \varphi)$$

Symbol	Name	Meaning
f	frequency (Hz $= \mathrm{s}^{-1}$)	oscillations per second; period $P = 1/f$
$\omega = 2\pi f$	angular frequency (rad s^{-1})	phase advance per second
A	amplitude	largest deviation from zero
φ	phase (rad)	where in its cycle the oscillation is at $t = 0$

Fourier's idea, for sampled data

Any N sampled values can be written **exactly** as a sum of cosines at the frequencies $f_k = k/T$, $k = 0, 1, \ldots, N/2$ – the oscillations with k whole periods in the record, period $P_k = T/k$ – each with its own amplitude A_k and phase φ_k.

The discrete Fourier transform

Problem: how do we find the amplitude and phase of each oscillation in the data?

$$X_k = \sum_{n=0}^{N-1} x_n \, e^{-2\pi i \, kn/N}, \qquad x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \, e^{2\pi i \, kn/N}$$

- X_k is complex: one number holds amplitude and phase
- Amplitude $A_k = 2|X_k|/N$, phase $\varphi_k = \arg X_k$ (for $0 < k < N/2$).
  Why: for one oscillation $x_n = A\cos(2\pi f_k t_n + \varphi)$ the sum gives $X_k = \frac{N}{2} A \, e^{i\varphi}$. N: the sum adds N terms; 2: a cosine is half at $+f_k$ and half at $-f_k$, $\cos\theta = (e^{i\theta} + e^{-i\theta})/2$, and only the $+f_k$ half is kept.
- Frequencies $f_k = k/T$: spacing $\Delta f = 1/T$

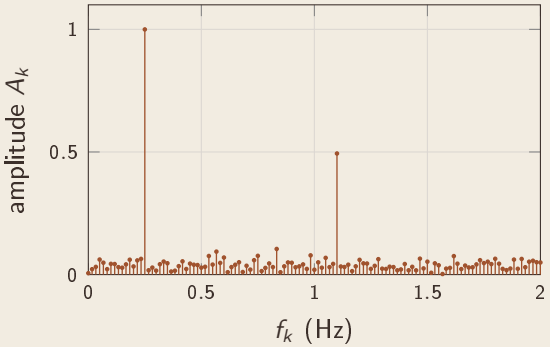
$e^{-2\pi ikn/N} = \cos(2\pi f_k t_n) - i\sin(2\pi f_k t_n)$: each X_k measures how much of the oscillation at f_k the series contains.

FFT: the same X_k, computed in about $N \log_2 N$ instead of N^2 operations (`numpy.fft.rfft`).

The spectrum of the running example

Problem: what are the oscillations in the running example (slide 5: amplitude 1 at 0.25 Hz, 0.5 at 1.1 Hz, plus noise)?

$N = 600$, $\Delta t = 0.1\,\text{s}$, $T = 60\,\text{s}$

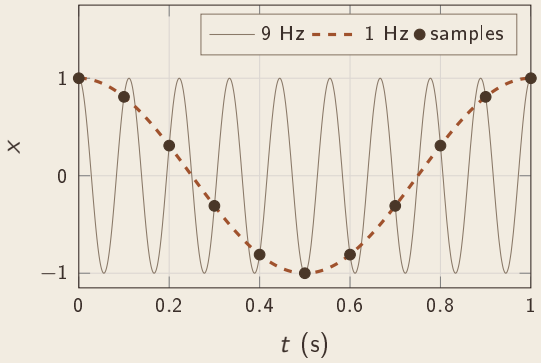


	true	from X_k
A at 0.25 Hz	1	1.000
A at 1.1 Hz	0.5	0.494
φ at 1.1 Hz	1.00	1.06
noise level	–	≈ 0.03

One value every $1/T = 0.017$ Hz. The two rhythms, hard to see in time, are two clear peaks with the right amplitudes.

Sampling: the fastest oscillation Δt can see

Problem: what is the fastest oscillation that samples taken every Δt can show?



Sampling at $f_s = 10$ per second: the samples of a 9-Hz cosine lie exactly on a 1-Hz cosine.

Nyquist frequency

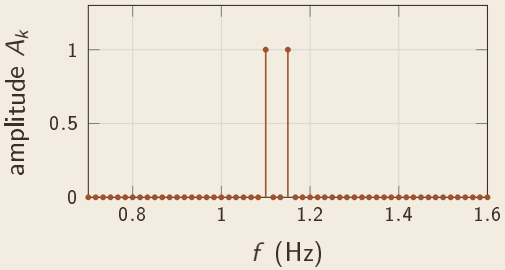
$f_N = \frac{f_s}{2} = \frac{1}{2\Delta t}$. A component above f_N appears at the wrong frequency $|f - mf_s|$, for the integer m that brings it below f_N: **aliasing**.

Cannot be undone after sampling: sample faster, or filter out fast components *before* sampling.

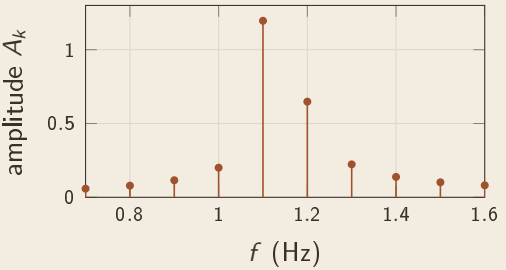
Record length: how close two frequencies can be

Problem: how close can two frequencies be and still be seen as two?

$T = 60\,\text{s}$: $\Delta f = 0.017\,\text{Hz}$



$T = 10\,\text{s}$: $\Delta f = 0.1\,\text{Hz}$

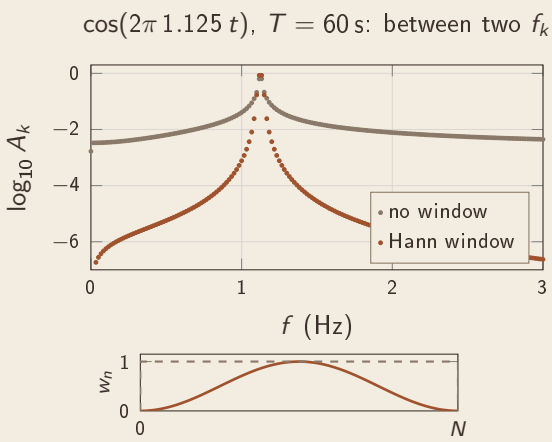


$\cos(2\pi\,1.10\,t) + \cos(2\pi\,1.15\,t)$. **Resolution** $\Delta f = 1/T$: to separate frequencies δf apart, record for $T \gtrsim 1/\delta f$ – here 20 s. The lowest frequency we can see is also about $1/T$.

Appending zeros (**zero padding**) gives a finer grid of values, not a finer resolution: still one peak. Zeros are an assumption (Lecture 4: padding is extrapolation).

Leakage: the record does not hold whole periods

Problem: what happens when the record does not hold a whole number of periods?



The DFT treats the record as one period of a periodic signal. A frequency off the grid makes a jump where the end meets the start, and its power **leaks** to all frequencies.

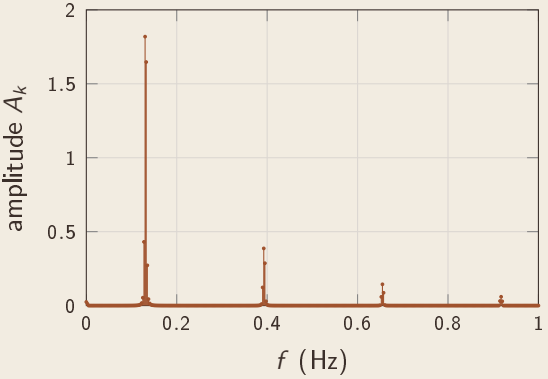
Window: multiply the data by $w_n = \sin^2(\pi n/N)$ (the **Hann window**), which goes smoothly to zero at both ends (dashed: no window). Then divide by $\sum w_n$ $(= N/2)$ instead of N, so that the amplitude is still A (sketch below the plot).

Peak height (true 1): 0.64 without, 0.85 with the window.

Cost: each peak about two Δf wide.

A peak is not always a separate oscillation

van der Pol, $T = 400\,\mathrm{s}$: one value every 0.0025 Hz



f (Hz)	multiple of f_0	amplitude
0.131	1	1.82
0.393	3	0.39
0.655	5	0.14
0.917	7	0.06

$f_0 = 1/P = 1/7.630\,\mathrm{s} = 0.131\,\mathrm{Hz}$. Even multiples: zero, because $x(t + P/2) = -x(t)$. Hann window (slide 17).

One oscillation, many peaks: the **harmonics** describe its non-sinusoidal shape.

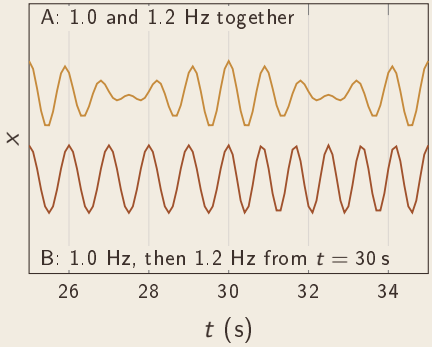
PART C

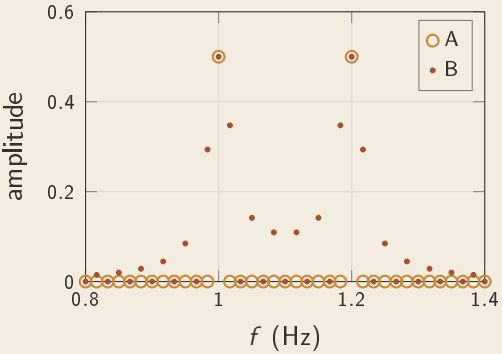
When the oscillations change in time

A spectrum averages over the whole record. What if the rhythms do not stay the same?

The spectrum says which frequencies, not when

Problem: two different signals – can their spectra tell them apart?



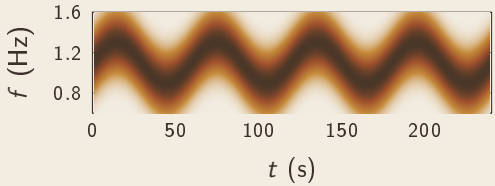


Both: peaks of 0.50 at 1.0 and 1.2 Hz ($T = 60$ s). Two oscillators, or one that changed its frequency? The amplitude spectrum alone cannot say; B's peaks are only wider, because each frequency lasted 30 s.

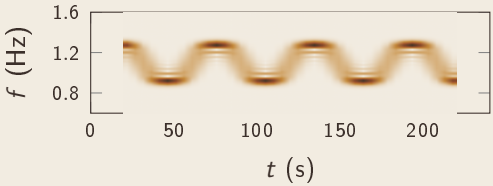
Short-time Fourier transform: the time–frequency trade-off

Problem: how do we see how the frequencies change in time?

Window $L = 3\,\mathrm{s}$



Window $L = 40\,\mathrm{s}$



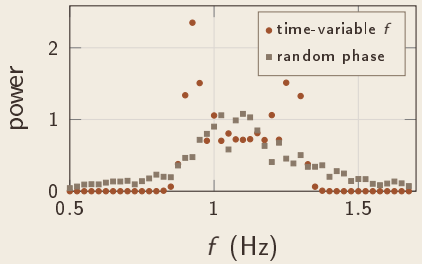
Short-time Fourier transform: the windowed DFT of the data within a window of length L, slid along the record. Resolution: about L in time and $1/L$ in frequency – one cannot have both.

L	$1/L$	What we see of $f(t) = 1.1 + 0.2\sin(2\pi t/60\,\mathrm{s})\,\mathrm{Hz}$
3 s	0.33 Hz	follows the changes, but blurred in frequency
12 s	0.08 Hz	both (next slide)
40 s	0.025 Hz	sharp in frequency, but the changes are smeared

A broad peak is not necessarily noise

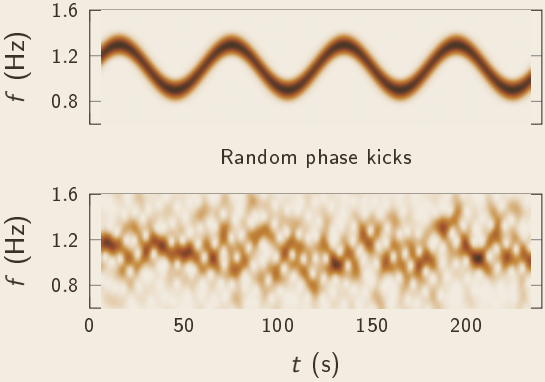
Problem: a broad peak – noise, or a rhythm whose frequency changes in time?

Average of the spectra of 40-s segments, $T = 240\,\mathrm{s}$



Cut the record into 40-s pieces, take the windowed DFT of each, and average: this reduces the random scatter of the spectrum.

Time-variable frequency (deterministic)



Top: $f(t) = 1.1 + 0.2\sin(2\pi t/60\,\mathrm{s})$ Hz, no noise. Bottom: 1.1 Hz, phase perturbed by independent random kicks. Both peaks are broad; only the time–frequency view shows why.

Summary

View	Tools	Check
Time	plot in order; amplitude, period, shape; running mean and spread; autocorrelation	stationarity; memory; events and drifts
Frequency	DFT via FFT; amplitude spectrum	aliasing ($f < 1/(2\Delta t)$); resolution $1/T$; leakage (window); padding is interpolation; harmonics
Time and frequency	short-time Fourier transform	window L: L in time against $1/L$ in frequency

Before you compute a spectrum: five questions

1. Is Δt short enough for the fastest component? *(aliasing: was it filtered before sampling?)*
2. Is T long enough for the slowest rhythm and the resolution I need? *($\Delta f = 1/T$)*
3. Is the series stationary, or do its rhythms change in time? *(time–frequency view)*
4. Did I use a window, and is padding showing more than the data contain? *(leakage)*
5. Are the peaks separate oscillations, harmonics, or leakage? *(check in time)*

Report Δt, T and window – and look again in time.

TO TAKE AWAY

A spectrum shows which rhythms, not when.

Δt sets the fastest, T the finest and slowest. Window, then look again in time: rhythms that change are not noise.